

kta Process Map

Proof of work for the generator-backed Sanskrit grammar archive

Repository purpose. This project is a long-term derivational index for deciding the kta form of a Sanskrit dhatu. Rules are stored once and reused through stable hyperlinks; examples can accumulate without duplicating explanation.

Decision flow

1. Dhatu identity	Resolve the exact Dhatupatha entry, gana, markers, meaning, preverb, and intended syntax.
2. Affix selection	Introduce kta and register it under the technical designation nistha.
3. Voice and meaning	Test ordinary passive/resultative interpretation and licensed kartari interpretation.
4. it decision	Apply the ardhadhatuka it system; do not infer set/anit/vet from gana alone.
5. Root operations	Apply root-specific substitutions, augment rules, and ordered phonology.
6. Nistha rules	Check 8.2.42ff for n-substitution, lexical outcomes, optionality, and prohibitions.
7. Sandhi and audit	Apply only triggered rules; record alternatives, sources, and unresolved questions.

Repository architecture

Preamble 1	Complete Dhatupatha registry, generated and stably anchored.
Preamble 2	Paninian sutra registry sorted numerically.
Chapters 1-11	Gana-specific derivation bars with local Rules Used anchors.
Cross-cutting	Sandhi, composition categories, irregular forms, and audit ledger.
Final proof	Compiled PDFs deposited under vyakaranamlikhitam/<project>/.

Initial audited rule backbone

Rule	Sutra key	Operational use
1.1.26	ktaktavatu nistha	Defines kta and ktavatu as nistha.
3.2.102	nistha	Introduces the nistha affixes in the completed-action domain.
3.4.72	gatyarthakarmaka...	Licenses kartari interpretation for motion roots, intransitives, and listed roots.
7.2.35	ardhadhatukasyed valadeh	Provides it before a val-initial ardhadhatuka affix, subject to exceptions.
8.2.42	radabhyam nisthato nah...	Begins central nistha substitutions after roots ending in r or d.
8.2.51	susah kah	Special k outcome for sus.
8.2.52	paco vah	Special v outcome for pac.
8.2.56	nudavidondatraghrahri...	Optional treatment for listed roots.
8.2.57	na dhyakhyapmurchimadam	Blocks the inherited operation for listed roots.
8.4.55	khari ca	Records a relevant final devoicing environment in tripadi.

Stable identifier examples

root: DHATU-01-0001
global rule: AS-8-2-42
local use: G01-AS-8-2-42-U01
derivation: G01-DHATU-01-0001-KTA
sandhi use: G01-SANDHI-AS-8-4-55-U01

Status. This is proof of implemented architecture, not a claim that every Sanskrit root has already received a fully audited kta derivation. The framework separates generated coverage from philologically verified coverage.